

FurFeel - ML Training Strategy (N = 3 Dogs)

When building a physiological machine learning model with only 2 to 3 unique subjects, the primary risk is Overfitting. If not handled correctly, the Random Forest will not learn what 'stress' looks like - it will simply memorize the personal biological quirks of Dog A and Dog B. If you put the collar on a new Dog C, the model will fail.

To successfully train a generalized model with only 3 dogs, you must implement the following 4 rules:

Rule 1: Strict Baseline Normalization

You cannot feed raw numbers to the AI.

Example: Dog A (a Chihuahua) has a resting Heart Rate (HR) of 110 bpm. Dog B (a Great Dane) has a resting HR of 60 bpm. If you feed raw HR to the Random Forest, it will assume 'high numbers mean stress' and constantly flag the Chihuahua as stressed, even when asleep.

Solution: Every feature fed to the AI must be a ratio relative to the dog's personal baseline. Instead of feeding 'HR = 120', feed 'HR_Ratio = 1.2x'. This completely masks the dog's breed and size from the AI, forcing it to look purely at physiological deviation.

Rule 2: Leave-One-Subject-Out (LOSO) Validation

Standard ML tutorials teach an '80/20 Random Split' (taking 80 percent of data for training and 20 percent for testing). With 3 dogs, this is disastrous. An 80/20 random split puts data from Dog A in both the training set AND the testing set. The AI will score 99 percent accuracy because it is being tested on a dog it has already seen.

Solution: You must use Leave-One-Subject-Out Cross-Validation (LOSO-CV). If you have Dog A, Dog B, and Dog C:

1. Train the model using ONLY data from Dog A and Dog B.
2. Test the model using ONLY data from Dog C.

If the Random Forest maintains high accuracy on Dog C, you know for a fact that it has successfully learned the general signs of stress, rather than memorizing Dog A and B.

Rule 3: Limit Model Complexity

A standard Random Forest uses deep decision trees (max_depth=None). A deep tree is smart enough to create highly specific rules to perfectly memorize your 3 dogs.

Solution: You must intentionally 'dumb down' the AI. By setting 'max_depth=3' or 'max_depth=4' in Scikit-Learn, you prevent the trees from growing deep enough to memorize individual dogs. The model will be forced to find broad, generalizable patterns (e.g., 'If HR_Ratio > 1.4 AND Motion > 0.5, then Stress').

Rule 4: Data Augmentation (Synthetic Data)

To prevent the AI from overfitting to the limited data, you can artificially multiply your dataset.

Solution: Take the telemetry data you collected for Dog A and Dog B, and 'jitter' it. Create a python script that copies a row of data, adds +/- 2 percent random noise to the Heart Rate, adds +/- 5 percent noise to the Respiratory Rate, and saves it as a new row. This simulates what the telemetry might look like if the dog was slightly larger or smaller, artificially expanding your 3 dogs into dozens of 'synthetic' dogs.